

Finance Credit Follow-Up Email Agent

Enterprise-style AI-powered platform for finance credit follow-up operations with deterministic workflow governance, constrained AI communication, human approval controls, dry-run execution, and LangGraph-based orchestration tracing.

Business Problem: Finance teams need consistent, auditable, and safe overdue payment follow-up workflows. Manual processes are hard to scale, difficult to govern, and often lack operational traceability.

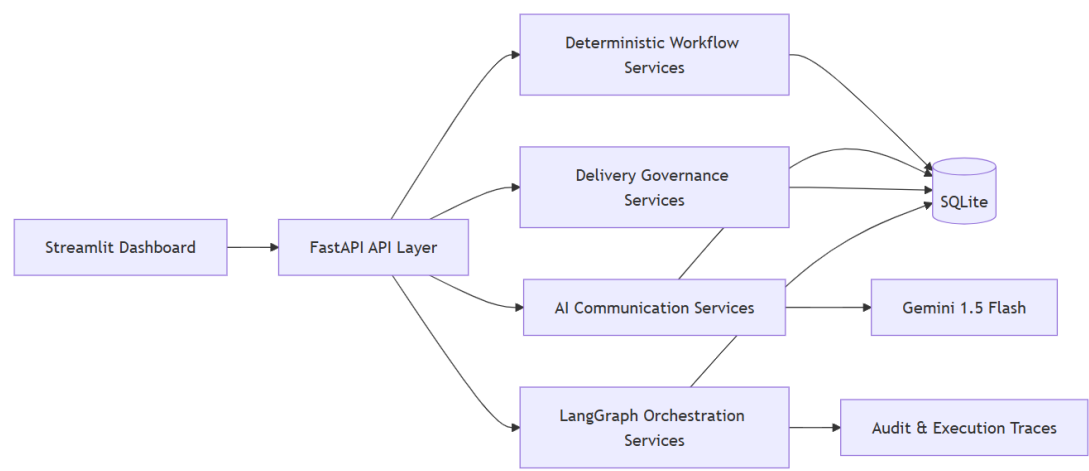
Key Features:

- Deterministic overdue and escalation workflow engine
- Stage-aware AI email generation with structured output validation
- Prompt-injection mitigation and forbidden language checks
- Human approval workflow before any delivery execution
- Dry-run delivery simulation (default-safe mode)
- LangGraph orchestration with pause/resume support
- Full audit trail for workflow, AI, delivery, and orchestration events
- Operations dashboard in Streamlit

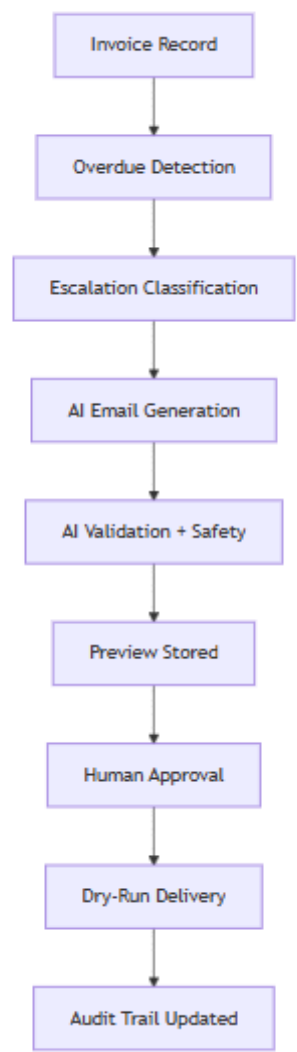
Architecture Overview

- backend/app/api: FastAPI route layer
- backend/app/services: business/application services
- backend/app/workflows: deterministic workflow engine + LangGraph orchestration
- backend/app/ai: AI provider abstraction, prompts, validation, sanitization
- backend/app/delivery: delivery provider abstraction and dry-run provider
- backend/app/models: ORM and Pydantic schemas
- frontend/streamlit_app.py: operations dashboard

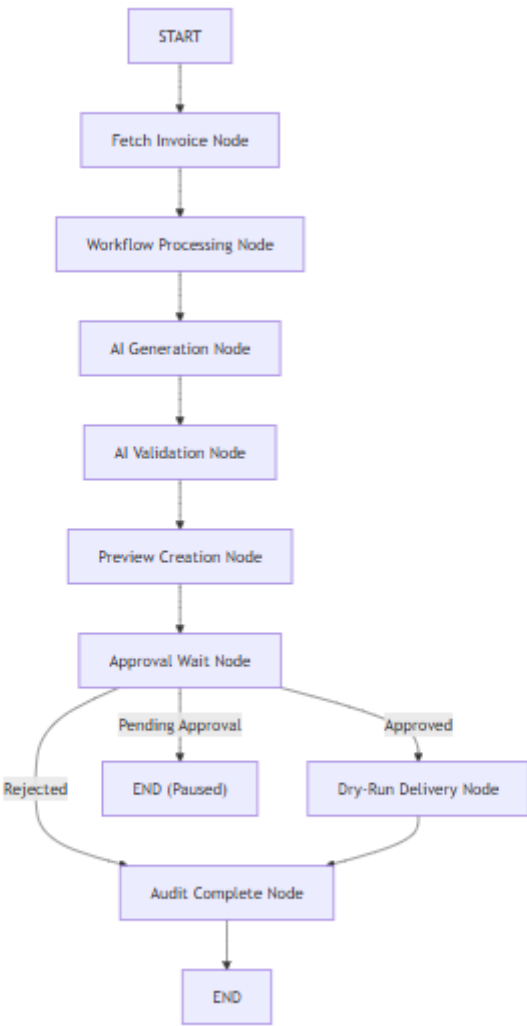
System Architecture



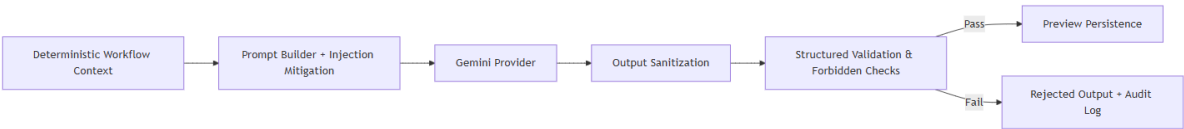
Workflow Lifecycle Diagram



LangGraph Orchestration Flow



AI Governance Pipeline



Workflow Lifecycle

- Invoice ingestion
- Overdue detection
- Escalation classification
- AI email generation (structured output)
- AI validation and safety checks
- Preview creation
- Human approval/rejection
- Dry-run delivery execution
- Audit and orchestration trace updates

LangGraph Orchestration

- Node-based stateful execution over existing services
- Approval interruption point (PAUSED_APPROVAL)
- Resume workflow support after approval decision
- Node transition history with durations and status

Endpoints:

- `POST /orchestration/run/{invoice_id}`
- `POST /orchestration/resume/{invoice_id}`
- `GET /orchestration/trace/{invoice_id}`
- `GET /orchestration/status/{invoice_id}`

AI Safety and Governance

- Externalized system/stage prompt files
- Structured output schema (subject, email_body, tone_used, escalation_stage)
- Hallucination checks against deterministic invoice context
- Forbidden phrase detection (threats, fabricated legal/penalty claims)
- Sanitization and normalization before validation acceptance
- Rejected outputs are logged and never advanced for delivery

Human Approval Workflow

Delivery statuses:

- GENERATED
- PENDING_APPROVAL
- APPROVED

- REJECTED
- DRY_RUN_SENT
- SENT
- FAILED

Safety guardrails:

- Unapproved emails cannot be sent
- Rejected content cannot be sent
- Duplicate dry-run sends are blocked

Tech Stack

- ❖ Frontend: Streamlit
- ❖ Backend: FastAPI
- ❖ Orchestration: LangGraph
- ❖ LLM: Gemini 1.5 Flash
- ❖ Database: SQLite + SQLAlchemy
- ❖ Validation: Pydantic
- ❖ Data processing: pandas

Setup Instructions

1. Create and activate a virtual environment.
2. Install dependencies: *pip install -r requirements.txt*
3. Copy environment template: *copy .env.example .env*
4. Set OPENAI_API_KEY in .env.

Run Locally

Backend:

```
python -m uvicorn app.main:app --app-dir backend --reload
```

Frontend:

```
python -m streamlit run frontend/streamlit_app.py
```

Optional frontend virtual environment (recommended on Windows):

```
python -m venv frontend/.venv
```

frontend/.venv/Scripts/python.exe -m pip install -r frontend/requirements.txt

frontend/.venv/Scripts/python.exe -m streamlit run frontend/streamlit_app.py

Or use helper scripts:

- [scripts/run_backend.ps1](#)
- [scripts/run_frontend.ps1](#)
- [scripts/run_all.ps1](#)

Single-command run (bootstraps venvs if needed):

powershell -ExecutionPolicy Bypass -File scripts/run_all.ps1

Environment Variables

See [.env.example](#) for full list.

Important variables:

- DATABASE_URL
- BACKEND_URL
- OPENAI_API_KEY
- OPENAI_MODEL_NAME
- OPENAI_TIMEOUT_SECONDS
- OPENAI_MAX_RETRIES

API Overview

Core:

- */invoices/**
- */workflows/**
- */audit*

AI:

- */ai/generate/{invoice_id}*
- */ai/generate-overdue-batch*
- */ai/generated-preview/{invoice_id}*

Delivery:

- */delivery/approve/{invoice_id}*
- */delivery/reject/{invoice_id}*
- */delivery/dry-run-send/{invoice_id}*
- */delivery/regenerate/{invoice_id}*
- */delivery/status/{invoice_id}*
- */delivery/pending-approvals*
- */delivery/summary*

Orchestration:

- */orchestration/run/{invoice_id}*
- */orchestration/resume/{invoice_id}*
- */orchestration/trace/{invoice_id}*
- */orchestration/status/{invoice_id}*

Streamlit Dashboard Overview

- Dashboard metrics and workflow processor
- Workflow stage filtering and legal escalation highlighting
- AI preview generation and validation indicators
- Delivery governance queue and status controls
- Orchestration run/resume/status/trace timeline viewer
- Audit log explorer

Screenshots:

Dashboard

Navigation

Go to

Dashboard

Invoice Upload

Invoice CSV

Drag and drop file here

Limit 200MB per file • CSV

Browse files

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Upload Invoices

Backend Connectivity

Test Health Endpoint

Deploy

Finance Credit Follow-Up Email Agent

Status: Deterministic workflow + hardened AI communication layer

Default Delivery Mode: DRY_RUN

Operations Snapshot

Total Invoices

6

Overdue Invoices

5

Escalations

5

Pending

6

Workflow Processing

Process Overdue Invoices

Recent Invoices

id	invoice_id	client_name	client_email	amount_due	due_date	follow_up_count	current_stage	payment_status
4	INV-1003	Contoso Retail	finance@contoso.com	980.00	2026-04-05	3	LEGAL_ESCALATION	PENDING
1	INV-2001	Blue Ocean Ltd	billing@blueocean.com	1500.50	2026-04-20	0	STAGE_3	PENDING
6	LIVE-RESEND-002	Vipin Yadav	vipinyadav.genaibu@gmail.com	1599.00	2026-04-20	0	STAGE_3	PENDING
2	INV-1001	Apex Manufacturing	billing@apexmfg.com	1250.00	2026-04-25	1	STAGE_3	PENDING
5	FINAL-E2E-002	Final E2E 002	final.e2e.002@example.com	1250.00	2026-04-30	0	STAGE_2	PENDING

Invoices

Navigation

Go to

Invoices

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Default Delivery Mode: DRY_RUN

Invoice List

API-driven data view

id	invoice_id	client_name	client_email	amount_due	due_date	follow_up_count	current_stage	payment_status
4	INV-1003	Contoso Retail	finance@contoso.com	980.00	2026-04-05	3	LEGAL_ESCALATION	PENDING
1	INV-2001	Blue Ocean Ltd	billing@blueocean.com	1500.50	2026-04-20	0	STAGE_3	PENDING
6	LIVE-RESEND-002	Vipin Yadav	vipinyadav.genaibu@gmail.com	1599.00	2026-04-20	0	STAGE_3	PENDING
2	INV-1001	Apex Manufacturing	billing@apexmfg.com	1250.00	2026-04-25	1	STAGE_3	PENDING
5	FINAL-E2E-002	Final E2E 002	final.e2e.002@example.com	1250.00	2026-04-30	0	STAGE_2	PENDING
3	INV-1002	Northwind Traders	ap@northwind.com	5400.00	2026-05-20	0	STAGE_1	PENDING

Current view: Invoices

Workflows

Navigation

Go to

Workflows

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Upload Invoices

Backend Connectivity

Test Health Endpoint

Deploy

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Status: Deterministic workflow + hardened AI communication layer

Default Delivery Mode: DRY_RUN

Workflow Overview

Deterministic Escalation Engine

Total Invoices

Overdue Invoices

Escalations

Pending

6

5

5

6

Filter by Escalation Stage

STAGE_1

invoice_id	client_name	client_email	amount_due	due_date	overdue_days	escalation_stage	payment_status	follow_up_count
INV-1002	Northwind Traders	ap@northwind.com	5400.00	2026-05-20	0	STAGE_1	PENDING	0

Legal Escalations

Manual Review Required

Legal escalation requires immediate review.

AI Preview

Navigation

Go to

AI Previews

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Upload Invoices

Backend Connectivity

Test Health Endpoint

Deploy

Finance Credit Follow-Up Email Agent

Status: Deterministic workflow + hardened AI communication layer

Default Delivery Mode: DRY_RUN

AI Email Preview Panel

Structured + Validated AI Output

Select overdue Invoice

INV-1003

Generate Preview for Selected Invoice

Fetch Latest Stored Preview

Batch Generation (Overdue)

Batch size

10

1

50

Generate Batch Previews

Current view: AI Previews

Delivery

Navigation

Go to

Delivery

Invoice Upload

Invoice CSV

Drag and drop file here

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Browse files

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Upload Invoices

Backend Connectivity

Test Health Endpoint

Deploy

Default Delivery Mode: DRY_RUN

Human Approval Gate Enabled

Delivery Governance

Delivery summary counts

```
{  "counts": {    "GENERATED": 0,    "PENDING_APPROVAL": 0,    "APPROVED": 0,    "REJECTED": 0,    "DRY_RUN_SENT": 3,    "SENT": 0,    "FAILED": 1  }}
```

No pending approvals.

Delivery action invoice

INV-1003

Approve

Dry-Run Send

Regenerate

Reject reason

Needs tone adjustment.

Reject

Orchestration

Navigation

Go to

Orchestration

Invoice Upload

Invoice CSV

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Upload Invoices

Backend Connectivity

Test Health Endpoint

Deploy

Default Delivery Mode: DRY_RUN

Traceable Stateful Coordination

Finance Credit Follow-Up Email Agent

Status: Deterministic workflow + hardened AI communication layer

LangGraph Orchestration

Orchestration invoice

CSV-DEMO-004

Run Workflow

Resume Workflow

Get Status

Get Trace

Execution Timeline

No trace events yet.

Current view: Orchestration

Audit

Navigation

Go to

Audit Logs

Invoice Upload

Invoice CSV

Drag and drop file here

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Browse files

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Upload Invoices

Backend Connectivity

Test Health Endpoint

Deploy

Finance Credit Follow-Up Email Agent

Status: Deterministic workflow + hardened AI communication layer

Default Delivery Mode: DRY_RUN

Audit Logs

Logs to fetch

10110500

id	invoice_id	action_type	status	timestamp	metadata_json
96	INV-1003	AI_GENERATION_ATTEMPT	SUCCESS	2026-05-10T15:13:57.158723	{"modelName":"gemini-2.5-flash-lite","prompt_version":"v1.1","stage_
95	INV-1003	AI_GENERATION_ATTEMPT	STARTED	2026-05-10T15:13:54.775012	{"modelName":"gemini-2.5-flash-lite","prompt_version":"v1.1","stage_
94	LIVE-RESEND-002	DELIVERY_SEND	FAILED	2026-05-10T14:55:23.611628	{"error":"Resend API error (403): {"statusCode":403,"message":"The
93	LIVE-RESEND-002	DELIVERY_APPROVAL	APPROVED	2026-05-10T14:55:22.409384	{"delivery_mode":"DRY_RUN"}
92	LIVE-RESEND-002	DELIVERY_SEND	FAILED	2026-05-10T14:52:24.483629	{"error":"Resend API error (403): {"statusCode":403,"message":"The
91	LIVE-RESEND-002	DELIVERY_APPROVAL	APPROVED	2026-05-10T14:52:23.997106	{"delivery_mode":"DRY_RUN"}
90	LIVE-RESEND-002	DELIVERY_SEND	FAILED	2026-05-10T14:51:55.826500	{"error":"Resend API error (403): {"statusCode":403,"message":"The
89	LIVE-RESEND-002	DELIVERY_APPROVAL	APPROVED	2026-05-10T14:51:54.642954	{"delivery_mode":"DRY_RUN"}
88	LIVE-RESEND-002	AI_GENERATION_ATTEMPT	SUCCESS	2026-05-10T14:51:54.522857	{"modelName":"gemini-2.5-flash-lite","prompt_version":"v1.1","stage_
87	LIVE-RESEND-002	AI_GENERATION_ATTEMPT	STARTED	2026-05-10T14:51:52.753759	{"modelName":"gemini-2.5-flash-lite","prompt_version":"v1.1","stage_

Current view: Audit Logs

Deployment Readiness

- Dockerfile: [Dockerfile](#)
- Docker Compose: [docker-compose.yml](#)

Run with Docker:

docker compose up --build

Demo

- Demo dataset: [data/demo/demo_invoices.csv](#)
- Demo dataset: [data/demo/test_invoice_data.csv](#)
- Captured demo outputs (CSV + JSON): docs/outputs/
- AI preview outputs: docs/outputs/output_aipreview/
- Workflow stage outputs: docs/outputs/output_workflow/

Future Improvements

- SMTP provider integration (with approval gate retained)
- Persistent orchestration state store (Redis/Postgres)
- Role-based access controls
- Notification and SLA monitoring
- CI pipeline for automated regression tests

Technical Stack and Decision Log

Mandatory Technical Disclosures

LLM Chosen

- Model: *Gemini 1.5 Flash*
- Provider: *Google Gemini API*
- Version/Identifier used in app config: *gemini-1.5-flash*
- Why this model:
 - Lower-latency generation suitable for operational follow-up workflows
 - Cost-effective for repeated preview generation and batch usage
 - Adequate instruction-following for structured JSON-style outputs
 - Good fit for controlled communication tasks with strict validation gates

Agent Framework

- Framework: *LangGraph (version pinned in requirements: 0.2.39)*
- Architecture style:
 - Stateful orchestration graph over deterministic service modules

- Plan-and-execute style node sequencing with explicit interruption points
- Single-orchestrator pattern (not multi-agent)
- Flow artifacts:
 - Graph lifecycle and node flow are documented in:
 - *docs/architecture_diagrams.md* (LangGraph orchestration flow)
 - *docs/demo_flow.md*

Prompt Design

- Prompt structure:
 - Externalized prompts in *backend/app/ai/prompts/*
 - One system prompt + stage-specific policy prompts
- Key system prompt:
 - File: *backend/app/ai/prompts/system_prompt.txt*
 - Defines enterprise finance assistant role, safe communication scope, and forbidden behaviors
- Guardrails applied:
 - Prompt injection pattern sanitization for user-controlled fields
 - Deterministic context injection from workflow data only
 - Structured output schema validation via Pydantic
 - Forbidden phrase and unsafe language detection

Security Risk Mitigation

Risk	Description	Mitigation Strategy
Prompt Injection	Malicious input manipulating agent behaviour	Input sanitization for customer-controlled fields, structured output schemas, strict validation and reject on fail behavior
API Key Exposure	LLM/email API keys leaked in code	.env + settings loader, no hardcoded keys, .env excluded via .gitignore, recommend secrets manager in production
Unauthorised Access	Anyone triggering the agent endpoint	Add API auth (API key/OAuth) before external exposure, apply rate limiting and gateway protections in deployment
Email Spoofing	Emails appearing from wrong sender	SPF/DKIM/DMARC with verified sender domain for real sending phase; keep dry-run mode as default in current testing phase